

AI Prompts Used

AI-Powered Invoice Extras Extraction

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1. Purpose

The Gemini prompt converts unstructured logistics email text into structured invoice-extra data. It is intentionally combined with a separate deterministic validation layer and final human approval/rejection.

2. Structured Output Requested

The AI returns: file reference, container number(s), positive customer additional charges, per-charge confidence, overall confidence, a Needs Review flag, and a short review reason.

3. Core Extraction Prompt

You are an accounting and logistics document extraction assistant.

Extract invoice-extra information from the email below.

IMPORTANT RULES:

1. Extract the file reference if present.
Typical format: SIM-0706-26 or SIM-0706-2026.
2. Extract all valid container numbers.
Typical format: four letters followed by seven digits.
3. Extract ONLY customer additional charges.
4. Ignore amounts equal to zero.
5. Do not invent any amount.
6. If quoted historical email text appears, prefer the latest/current message.
7. Do not count a written TOTAL as a separate charge if individual charges are already listed.
8. Standardize obvious spelling variations:
FERRI -> FERRY
DEMURAGE -> DEMURRAGE
9. If the email is ambiguous, incomplete or the relationship between charges and containers is uncertain, set needs_review = true.
10. Confidence must reflect actual certainty. Do not automatically assign 100.

EMAIL:

{email_text}

4. Safety / Control Logic After AI

After Gemini returns structured output, the application removes any charge with amount ≤ 0 . The independent validation module then checks mandatory identifiers, low confidence, AI uncertainty and duplicate charges. The human reviewer retains the final Approve / Reject decision.

5. Prompt Design Rationale

- The prompt limits extraction to customer-recoverable additional charges.
- It explicitly prohibits invented amounts.
- It prefers the current message over quoted historical email content.
- It avoids double-counting a written total when individual charges are present.

- It asks the model to expose uncertainty rather than silently treating ambiguous text as certain.

6. Security Note

The Gemini API key is not part of this prompt and must not be included in the project submission. The prototype reads GEMINI_API_KEY from the Windows environment.

7. Limitation

Prompt behavior can vary across model versions and unseen email formats. The prompt is therefore not used as the sole control; deterministic validation and human review remain part of the design.